

AWS Project: Deploy Website on EC2 & Configure Security Groups: -

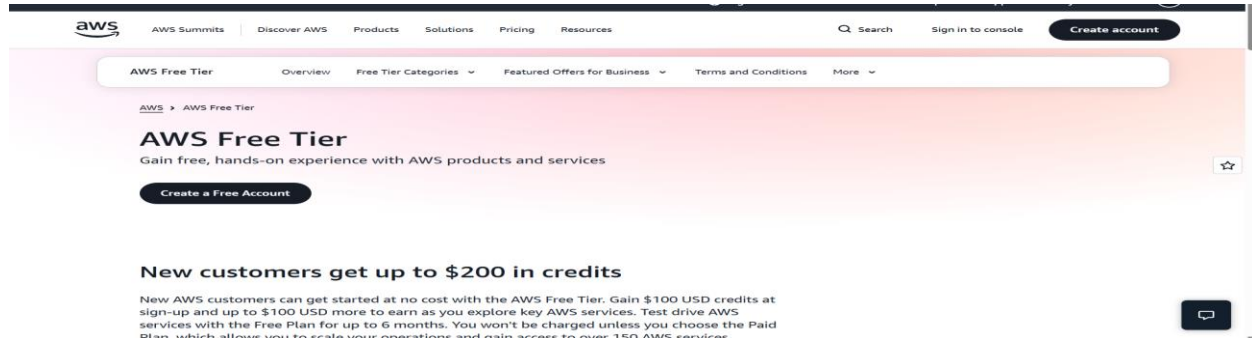
Project Objective: Launch an EC2 instance on AWS and host a simple website accessible from the internet.

AWS Services Used

Service	Purpose
Amazon EC2	Virtual server for website hosting
Security Group	Firewall rules for traffic control
Public IP	Internet accessibility
Apache HTTP Server	Web server to host website

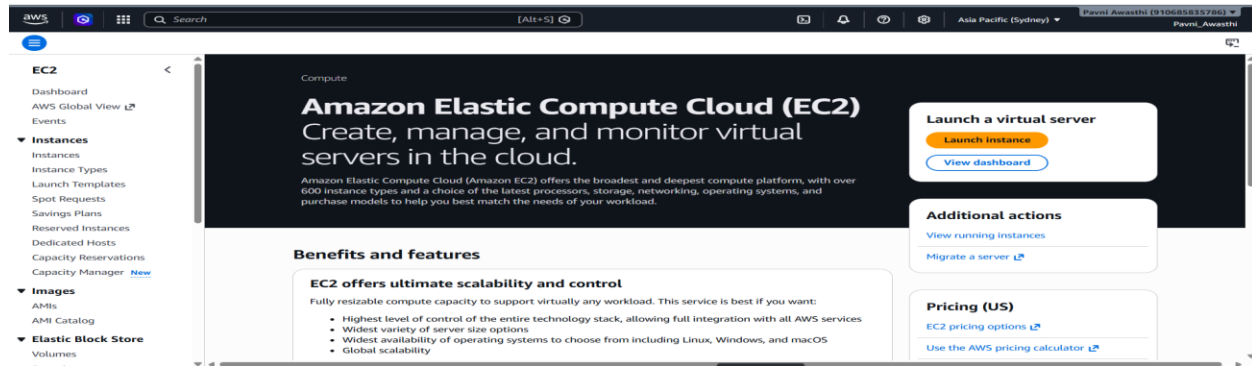
Step1: - Login to AWS Console

Open AWS Management Console and login using an AWS account.



Step2: - Open EC2 Dashboard

Search for EC2 service and click 'Launch Instance'.



Step3: - Configure EC2 Instance

Choose Amazon Linux 2 AMI, t2. micro instance type and create /select key pair.

EC2 > Instances > Launch an instance

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-080beda09284d47e2 (64-bit (x86), uefi-preferred) / ami-06ef0823cb7e4f283 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260526.0 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-080beda09284d47e2

Publish Date

2026-05-21

Username

ec2-user

Verified provider

Instance type

Info | Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0167 USD per Hour

On-Demand RHEL base pricing: 0.042 USD per Hour On-Demand Windows base pricing: 0.0224 USD per Hour

On-Demand SUSE base pricing: 0.0132 USD per Hour On-Demand Linux base pricing: 0.0132 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Launch an instance

Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Info

Name

my web server

Add additional tags

Application and OS Images (Amazon Machine Image)

Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mywebkey

Create new key pair

Step4: - Configure Security Group

EC2 > Security Groups > sg-0df61f77e12343f33 - launch-wizard-3 > Edit inbound rules

Edit inbound rules

Inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
-	SSH	TCP	22	Anywh...	0.0.0.0/0	Delete
-	HTTP	TCP	80	Anywh...	0.0.0.0/0	Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

Step5: - Launch EC2 Instance

Click on the launch instance and wait until the instance is running.

Instance summary for i-03ab0f070bfe74a14 (my web server) info

Updated less than a minute ago

Instance ID
i-03ab0f070bfe74a14

IPv6 address
-

Hostname type
IP name: ip-172-31-3-102.ap-southeast-2.compute.internal

Answer private resource DNS name
IPv4 (A)
32.236.188.179 [Public IP]

Auto-assigned IP address
32.236.188.179 [Public IP]

IAM role
-

IMDSv2
Required

Public IPv4 address
32.236.188.179 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-3-102.ap-southeast-2.compute.internal

Instance type
t3.micro

VPC ID
vpc-0df5e6dacc7e6e522

Subnet ID
subnet-022acba706c4e0511

Instance ARN
arn:aws:ec2:ap-southeast-2:910685835786:instance/i-03ab0f070bfe74a14

Private IPv4 addresses
172.31.3.102

Public DNS
ec2-32-236-188-179.ap-southeast-2.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendation
[Learn more](#)

Auto Scaling Group name
-

Managed
false

Step6: - Connect to EC2

Use EC2 Instance Connect or SSH command to connect to the server.

Connect info

Connect to an instance using the browser-based client.

Instance ID
i-03ab0f070bfe74a14 (my web server)

VPC ID
vpc-0df5e6dacc7e6e522

Security groups
sg-0df61f77e12343f33 (launch-wizard-3)

IAM role
-

EC2 Instance Connect | SSM Session Manager | SSH client | EC2 serial console

Instance ID
i-03ab0f070bfe74a14 (my web server)

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

Public IPv4 address
32.236.188.179

IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
ec2-user@ip-172-31-3-102 ~$
```

Step7: - Install Apache Web Server

Run commands to install Apache HTTP server.

sudo yum update -y

```

  ____  __
 / ___/  / /_  __
/ /   / __/ / / /
/ /___/ /_/_/ /_/_
/_____/

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-3-102 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
```

373 kB/s | 47 kB 00:00

sudo yum install httpd -y

```

[ec2-user@ip-172-31-3-102 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:00:34 ago on Thu Jun  4 13:18:47 2026.
Dependencies resolved.

Package                               Architecture      Version            Repository          Size
Installing:
httpd                                  x86_64            2.4.67-1.amzn2023.0.1  amazonlinux        46 k
Installing dependencies:
apr                                     x86_64            1.7.5-1.amzn2023.0.4  amazonlinux        129 k
apr-util                               x86_64            1.6.3-1.amzn2023.0.2  amazonlinux        13 k
apr-util-ldap                          x86_64            1.6.3-1.amzn2023.0.2  amazonlinux        19 k
generic-logos-httpd                   noarch            18.0.0-12.amzn2023.0.3  amazonlinux        12 k
httpd-core                             x86_64            2.4.67-1.amzn2023.0.1  amazonlinux        1.4 M
httpd-tools                             x86_64            2.4.67-1.amzn2023.0.1  amazonlinux        80 k
libbrotli                               x86_64            1.0.9-4.amzn2023.0.2  amazonlinux        315 k
mailcap                                noarch            2.1.49-3.amzn2023.0.3  amazonlinux        33 k
Installing weak dependencies:
apr-util-openssl                       x86_64            1.6.3-1.amzn2023.0.2  amazonlinux        15 k
mod_http2                              x86_64            2.0.39-1.amzn2023.0.1  amazonlinux        167 k
mod_lua                                x86_64            2.4.67-1.amzn2023.0.1  amazonlinux        59 k

Transaction Summary
Install 13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm 327 kB/s | 13 kB 00:00
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm          2.0 MB/s | 129 kB 00:00
```

```

(1/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm 327 kB/s | 13 kB 00:00
(2/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm          2.0 MB/s | 129 kB 00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm      1.9 MB/s | 97 kB 00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm 643 kB/s | 15 kB 00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 799 kB/s | 19 kB 00:00
(6/13): httpd-2.4.67-1.amzn2023.0.1.x86_64.rpm        1.0 MB/s | 46 kB 00:00
(7/13): httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch.rpm 431 kB/s | 12 kB 00:00
(8/13): httpd-core-2.4.67-1.amzn2023.0.1.x86_64.rpm    31 MB/s | 1.4 MB 00:00
(9/13): httpd-tools-2.4.67-1.amzn2023.0.1.x86_64.rpm   2.0 MB/s | 80 kB 00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm     10 MB/s | 315 kB 00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm       1.4 MB/s | 33 kB 00:00
(12/13): mod_http2-2.0.39-1.amzn2023.0.1.x86_64.rpm    6.6 MB/s | 167 kB 00:00
(13/13): mod_lua-2.4.67-1.amzn2023.0.1.x86_64.rpm      2.4 MB/s | 59 kB 00:00

Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : apr-1.7.5-1.amzn2023.0.4.x86_64                                1/1
  Installing     : apr-util-1.6.3-1.amzn2023.0.2.x86_64                        2/13
  Installing     : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64                  3/13
  Installing     : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64              4/13
  Installing     : mailcap-2.1.49-3.amzn2023.0.3.noarch                      5/13
  Installing     : httpd-tools-2.4.67-1.amzn2023.0.1.x86_64                  6/13
  Installing     : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                     7/13
  Running scriptlet: httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch             8/13
  Installing     : httpd-core-2.4.67-1.amzn2023.0.1.x86_64                   9/13
  Installing     : httpd-2.4.67-1.amzn2023.0.1.x86_64                       10/13
  Installing     : mod_http2-2.0.39-1.amzn2023.0.1.x86_64                    11/13
  Installing     : mod_lua-2.4.67-1.amzn2023.0.1.x86_64                      12/13
  Running scriptlet: httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch             13/13
  Verifying      : apr-1.7.5-1.amzn2023.0.4.x86_64                          1/13
  Verifying      : apr-util-1.6.3-1.amzn2023.0.2.x86_64                      2/13
  Verifying      : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64                 3/13
  Verifying      : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64              4/13
  Verifying      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          5/13
  Verifying      : httpd-2.4.67-1.amzn2023.0.1.x86_64                       6/13
  Verifying      : httpd-core-2.4.67-1.amzn2023.0.1.x86_64                  7/13
  Verifying      : httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch              8/13
  Verifying      : httpd-tools-2.4.67-1.amzn2023.0.1.x86_64                 9/13
  Verifying      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                    10/13
  Verifying      : mailcap-2.1.49-3.amzn2023.0.3.noarch                      11/13
  Verifying      : mod_http2-2.0.39-1.amzn2023.0.1.x86_64                    12/13
  Verifying      : mod_lua-2.4.67-1.amzn2023.0.1.x86_64                     13/13
```

```

Installing      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                      7/13
Running scriptlet: httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch             8/13
Installing      : httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch             8/13
Installing      : httpd-core-2.4.67-1.amzn2023.0.1.x86_64                   9/13
Installing      : mod_http2-2.0.39-1.amzn2023.0.1.x86_64                    10/13
Installing      : mod_lua-2.4.67-1.amzn2023.0.1.x86_64                      11/13
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          12/13
Installing      : httpd-2.4.67-1.amzn2023.0.1.x86_64                       13/13
Running scriptlet: httpd-2.4.67-1.amzn2023.0.1.x86_64                       13/13
Verifying      : apr-1.7.5-1.amzn2023.0.4.x86_64                          1/13
Verifying      : apr-util-1.6.3-1.amzn2023.0.2.x86_64                      2/13
Verifying      : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64                 3/13
Verifying      : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64              4/13
Verifying      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch          5/13
Verifying      : httpd-2.4.67-1.amzn2023.0.1.x86_64                       6/13
Verifying      : httpd-core-2.4.67-1.amzn2023.0.1.x86_64                  7/13
Verifying      : httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch              8/13
Verifying      : httpd-tools-2.4.67-1.amzn2023.0.1.x86_64                 9/13
Verifying      : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                    10/13
Verifying      : mailcap-2.1.49-3.amzn2023.0.3.noarch                      11/13
Verifying      : mod_http2-2.0.39-1.amzn2023.0.1.x86_64                    12/13
Verifying      : mod_lua-2.4.67-1.amzn2023.0.1.x86_64                     13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.2.x86_64          apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.67-1.amzn2023.0.1.x86_64
httpd-core-2.4.67-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.67-1.amzn2023.0.1.noarch      httpd-tools-2.4.67-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64        mailcap-2.1.49-3.amzn2023.0.3.noarch              mod_http2-2.0.39-1.amzn2023.0.1.x86_64
mod_lua-2.4.67-1.amzn2023.0.1.x86_64

Complete!
```

Step8: - Start Apache Services

Start and enable Apache Service using systemctl commands.

sudo systemctl start httpd and sudo systemctl enable httpd

```
[ec2-user@ip-172-31-3-102 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-3-102 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service -> /usr/lib/systemd/system/httpd.service.
```

Step 9: - Create website

Navigate to /var/www/html and create an index.html file.

```
[ec2-user@ip-172-31-3-102 ~]$ cd /var/www/html
```

```
[ec2-user@ip-172-31-3-102 html]$ sudo nano index.html
```

i-03ah0f070hfe74a14 (my web server)

GNU nano 8.3

index.html

Step 10: - Access website

Open browser using Public IPv4 address of EC2 instance

← → 🔍 Not secure 32.236.188.179 ☆ 🚫 ⓘ